The effectiveness of aluminium as a prosthetic in different conditions

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Background:

Aluminium alloys are commonly used in prosthetic components due to their high strength-to-weight ratio, corrosion resistance, and ease of machining. These properties allow prosthetic devices to remain lightweight while supporting the repeated loads generated during daily activities. Finite element analysis of hip prosthesis stems has reported peak stresses of approximately 95–101 MPa under static and dynamic loading conditions (El'Sheikh, et al., 2003), while experimental prosthetic socket testing has measured von Mises stresses around 34 MPa under functional loading (Mechi & Al-Waily, 2021). Mechanical testing is crucial to ensure the material is suitable for functioning under these stresses.

However, as humans live in different environments and have different lifestyles, their prosthetics experience different conditions, which may alter their mechanical behaviour and lifetime. Compression testing in different conditions provides stress–strain behaviour, yield strength, and Young's modulus to evaluate material suitability for load-bearing prosthetic applications in various environments.

Aim:

- To assess the suitability of aluminium as a load-bearing material for prosthetic applications
- To study the effect of different loading rates and cyclic compression on the alloy
- To evaluate its mechanical performance under temperature variations

Hypothesis:

- Aluminium is expected to exhibit predominantly elastic behaviour under compressive loading, with limited permanent deformation.
- Increasing the ramp rate is expected to increase the yield strength of aluminium, as faster loading may increase resistance to deformation.
- Under cyclic loading simulating prosthetic leg and arm use, aluminium is expected to demonstrate minimal strain accumulation and strong fatigue resistance.
- Lower temperatures (-18°C) are expected to increase stiffness, while higher temperatures (38°C) may slightly increase deformation under repeated loading.

Results

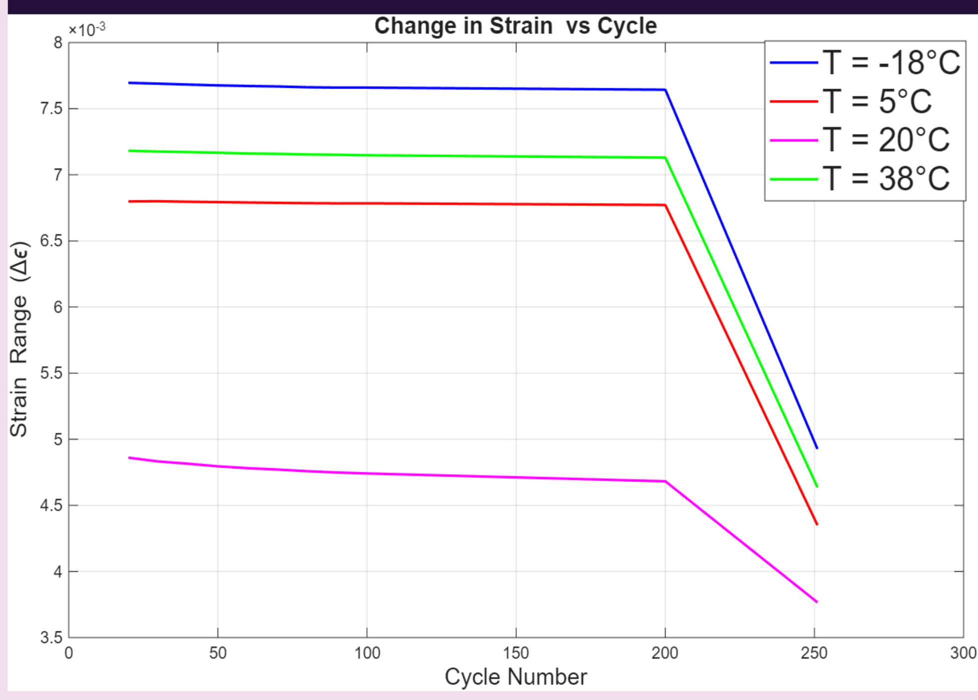


Figure 1: Variation of strain with cycle number during cyclic compression (Test 2)

- The strain range remains relatively constant, suggesting a stable cyclic deformation response.
- The 20 °C dataset exhibits the lowest strain range, suggesting a reduced deformation in operating temperatures. The more extreme temperatures (-18 °C and 38 °C) showed the greatest deformation, indicating the effect of temperature on the mechanical behaviour of aluminium.
- The similar shape of all curves indicates temperature influences deformation magnitude rather than cyclic behaviour.
- The drop near 200 cycles likely corresponds to the final unloading stage of the test, rather than material fatigue.
- Across all temperatures, the samples presented extremely small changes in strain, showing their suitability in high load baring conditions.

- The consistent separation between the two datasets indicates that increasing load increases the final deformation across all temperatures.
- Final deformation varies with temperature and applied load, with higher loads (Test 3) generally producing greater deformation than the lower load condition (Test 2).
- At extreme conditions, deformation is relatively higher under both loading conditions, suggesting the material experiences greater permanent deformation.
- The lowest deformation occurs at 20 °C, indicating improved dimensional stability at operational temperatures.

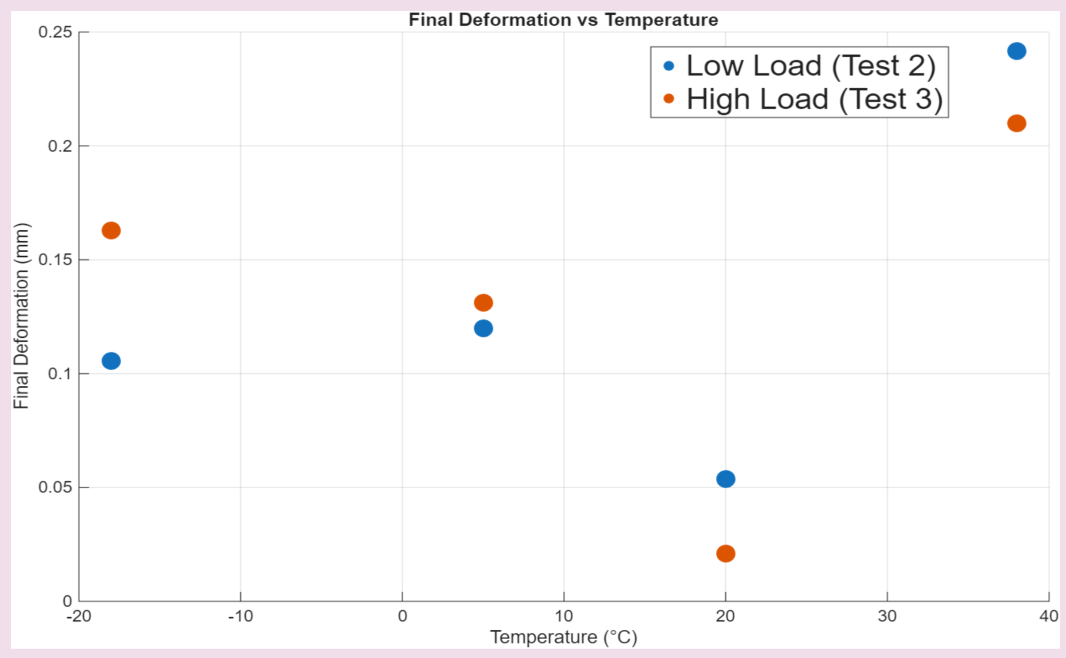


Figure 2: Final deformation against temperature during cyclic compression under low (Test 3) and high (Test 2) load

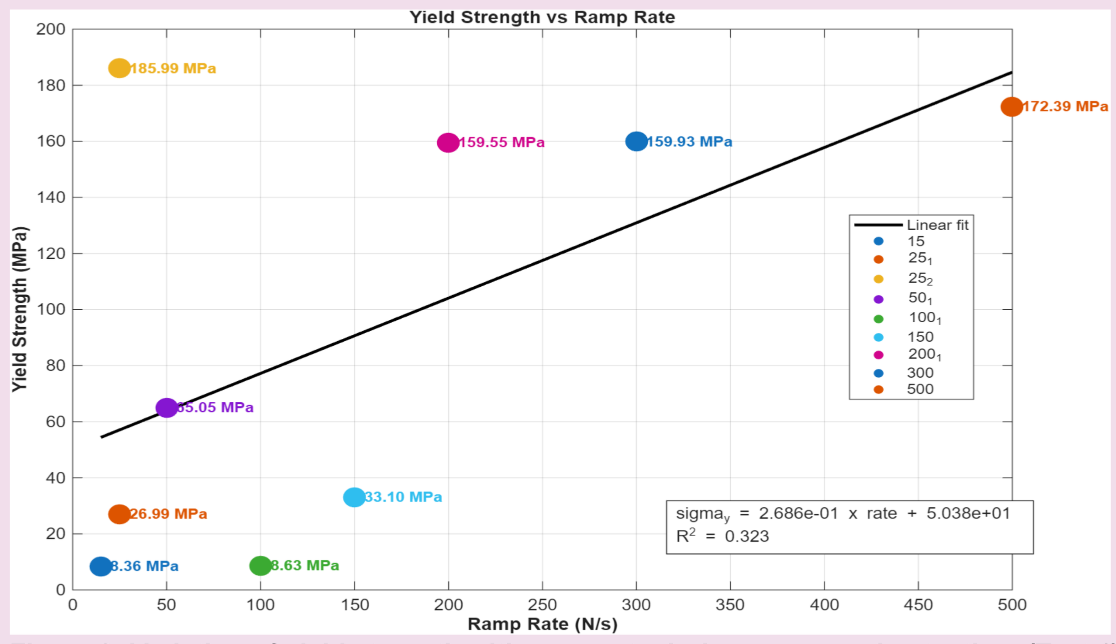


Figure 3: Variation of yield strength with ramp rate during compression testing. (Test 1)

- Yield strength generally increases with increasing ramp rate, indicating that the material exhibits strain-rate dependent behaviour.
- At low ramp rates (15–100 N/s), the yield strength remains relatively low, while at higher ramp rates (200–500 N/s), the yield strength increases significantly, indicating greater resistance to deformation under faster loading conditions.
- The positive slope of the linear fit confirms an overall trend of increasing yield strength with ramp rate, however, the relatively low R2 value indicates the relationship is moderate rather than strongly linear.

Experimental Design:

The following compression tests were conducted on aluminium samples, with a diameter of 3.98mm and a length of 8.44mm, using the Instron ElectroPuls E3000:

Test 1: Changing Ramp Rates	Test 2: Cyclic compression under higher force amplitude	Test 3: Cyclic compression under lower force amplitude
• Load to 2500N, then unloaded to 10N • For ramp rates 200N/s and 300N/s	• Compressed from 100N/s to 800N, then perform 250 cycles at 700N and 0.77Hz on blocks exposed to -18°C, 5°C, and 38°C for a week.	• Compressed from 110N/s to 330N, then 300 cycles at 320N and 0.5Hz, with blocks exposed to -18°C, 5°C, and 38°C for a week

Test 1, investigating whether loading rate affects the mechanical response of aluminium:

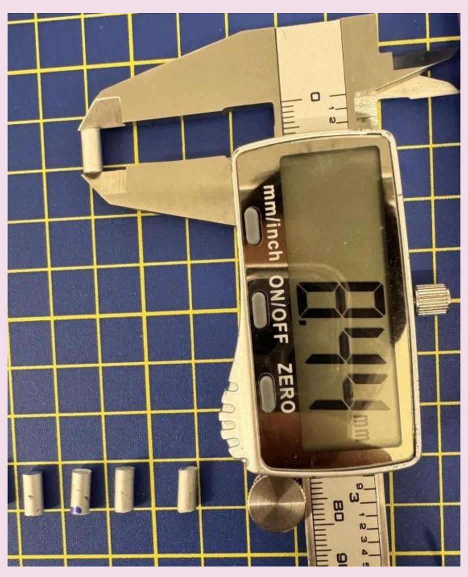
Prosthetic components face varying loading rates during activities, slower loading occurs during standing or weight shifting, while faster loadings occur during walking, running, or sudden movements. Tests at ramp rates of 200 N/s and 300 N/s assessed strain-rate sensitivity. A maximum load of 2500 N was chosen to evaluate worst-case conditions and ensure a safety margin.

Test 2, investigating the behaviour of aluminium in higher load-bearing conditions:

Lower-limb prostheses typically support 1–1.2 times body weight per step (Puthenpurackel, 2016). Aluminium blocks were subjected to 250 loading cycles at 700 N and 0.77 Hz, simulating a walking gait. Room-temperature experiments were taken from previous datasets, alongside the tested temperature conditions.

Test 3, investigating the behaviour of alumium in moderate load-bearing conditions:

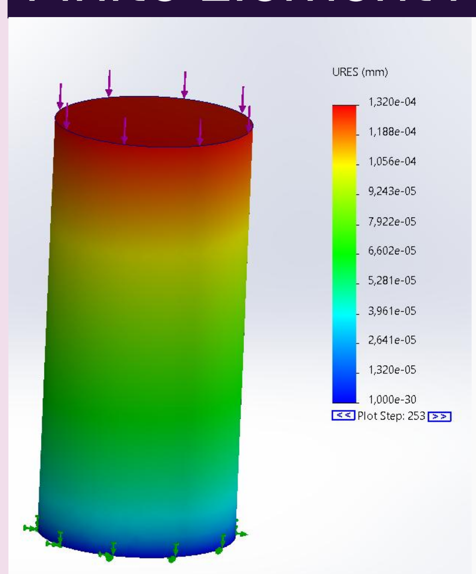
This test is similar to test 2, but designed for upper-limb prosthetics, which typically experience lower absolute loads, but often undergo higher numbers of repeated cycles during daily activities. Ensuring reliability for prosthetic arm components.



Evaluation of model and method:

- The additional dataset used aluminium blocks with different dimensions. The experimental method did not originally account for this, limiting comparability.
- Tests were conducted at room temperature, this led to partial temperature changes within the sample during the experiment.
- During the cyclic tests, the Instron machine did not record any data between the 100th and the 200th cycle, and the 200th and the 250th cycle. This loss of intermediate data limited fatigue analysis and strain evolution tracking.
- Cyclic loading was modelled as a sine wave of constant amplitude to represent walking. A real prosthetic will experience multi-axial forces (e.g. Torsion), and variable amplitudes during loading. (Taylor et al., 2020)

Finite Element Analysis:



Finite Element Analysis was conducted on SolidWorks, the results were then compared to our raw data. Full analysis can be viewed in our website by scanning the following QR code:

	Young's Modulus (E) /GPa	Yield Strenght/MPa
Theoretical	70	105-110
Test (200N/s)	6.41	159.93

Conclusion:

The experimental results show that:

- Aluminium maintains a relatively stable response during repeated loading cycles, with only small variations in strain, making it suitable to withstand the required loads.
- Deformation increases with higher applied loads, confirming that load magnitude plays an important role in the material's mechanical behaviour.
- Temperature also influences deformation, with slightly higher strain observed at more extreme temperatures compared with room temperature conditions. However, these differences are in the μm range, creating a negligible difference.

Overall, the results support the hypothesis that aluminium can maintain mechanical stability under varying loading rates, cyclic loading, and environmental conditions.

These findings suggest aluminium is a suitable lightweight material for load-bearing prosthetic components, although further testing under multi-axial loading and longer fatigue cycles would better replicate real-world prosthetic use.

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